

BINDU CHANDRA SHEKAR REDDY

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Professional Summary

Machine Learning Engineer and Software Developer with mid-level experience delivering scalable AI solutions and full-stack applications. Designed and deployed an end-to-end healthcare AI pipeline presented at IEEE SysCon, and optimized web APIs to cut response time by 30% while improving reliability. Skilled in transformer-based models, cloud-native architectures, and real-time inference pipelines. Aiming to apply this blend of ML and software engineering expertise to accelerate intelligent product development.

Skills

- **Programming Languages:** PHP, Python, Java, C, SQL, JavaScript
- **Frameworks & Technologies:** Node.js, React, Spring Boot, REST APIs
- **Machine Learning:** Transformers, CNN, ANN, XGBoost, NumPy, Pandas, Matplotlib, TensorFlow/PyTorch, Scikit-learn
- **Cloud & DevOps:** Docker, Kubernetes, Google Kubernetes Engine (GKE), AWS
- **Tools:** Git, VS Code, Google Colab, Xcode

Experience

Teleindia Networks Pvt. Ltd

Jun 2023 - Jul 2024

Web Developer

Bengaluru, India

- Developed and maintained responsive web applications using React, PHP and Node.js, applying CSS modules and Jest testing, which improved usability and cross-device performance as reflected in positive user feedback
- Implemented RESTful APIs and optimized MySQL queries, reducing average response time by 30%.
- Collaborated with designers and product teams, using React and PHP to deliver full-stack solutions that met business requirements and enabled on-time release of key features
- Integrated Java-based backend modules with Spring Boot, using Docker containers to ensure seamless data flow between client and server layers, which eliminated data synchronization errors
- Participated in incident management, debugging and resolving production issues to enhance reliability by 20%.

California State University, Sacramento

Aug 2025 - Present

Instructional Student Assistant

Sacramento

- Guided students in applying object-oriented programming and computer-graphics concepts using Java, helping them complete weekly coding assignments and deepening their ability to create interactive visualizations
- Supported project reviews and lab exercises using VS Code and Git, guiding students through animation, event handling, and software-design tasks, which resulted in clearer concepts and higher assignment scores

California State University Sacramento

Feb 2024 - Dec 2024

Teaching Assistant

Sacramento

- Assisted students in learning Java fundamentals, focusing on object-oriented design, debugging and coding best practices, which helped them write clearer and more reliable code.
- Conducted lab sessions and gave targeted feedback on assignments through collaborative tools such as Google Colab and Git, which strengthened students' programming skills and deepened their conceptual understanding

Nanorobotics Embed Technologies

Sep 2021 - Oct 2021

Machine Learning Intern

Bengaluru, India

- Developed and tested ANN models in Python with NumPy, Pandas and Matplotlib, achieving reliable predictions that supported downstream embedded analytics
- Programmed Python-based software modules for embedded devices and built ML inference pipelines, enabling real-time analytics for improved device performance.
- Presented findings on ML integration for embedded robotics to the engineering team, informing design decisions and advancing the project roadmap

Projects

NeuroFusionGPT – Multimodal Stress Detection System

- Built a multimodal deep learning model that predicts 5 stress levels by fusing EEG (brain signals) and ECG (heart signals) using a feature-level neural fusion architecture.
- Implemented Transformer-based EEG modeling (16 features) and MLP-based ECG analysis (188 features) to learn physiological stress patterns from 95K+ signal samples.
- Developed an end-to-end ML pipeline including preprocessing, feature scaling, multimodal fusion, inference, and evaluation using accuracy, precision, recall, F1-score and confusion matrix.
- Integrated an AI wellness feedback module that generates personalized stress-management guidance based on predicted stress levels; trained models in Google Colab (GPU) and deployed locally in Python.

ML-based Medical Diagnostics for Infectious Diseases

- Published at IEEE International Systems Conference (SysCon 2026), focusing on scalable ML-based diagnostic systems.
- Developed a supervised ML pipeline using epidemiological data (263K+ samples) for COVID-19 prediction.
- Engineered clinical and temporal features (comorbidity count, symptom-to-admission interval) to improve predictive performance.
- Built and optimized an XGBoost model achieving 0.80 ROC-AUC and 0.73 Recall for high-risk case detection.
- Designed a robust preprocessing pipeline to handle noisy, imbalanced real-world healthcare data.

IoT-Based Health Monitoring System with Edge Computing

- Developed an iOS frontend for real-time health tracking via connected IoT devices.
- Integrated edge computing for local data processing to reduce latency and improve performance.
- Containerized services using Docker and deployed on Google Kubernetes Engine (GKE) for scalable backend support.

Plant Leaf Disease Detection Using CNN

- Engineered a CNN model using transfer learning (VGG16, ResNet50), achieving ~95% classification accuracy on plant disease datasets.
- Processed and visualized data using Python libraries (NumPy, Pandas, Matplotlib).
- Delivered a real-time classification system enabling early disease detection and preventive action for farmers.

Publications

- Bindu Chandra Shekar Reddy et al..ML-based Medical Diagnostics System for Infectious Diseases (Case Study: COVID-19).*Published at IEEE International Systems Conference (SysCon 2026). Developed a machine learning-based diagnostic system using epidemiological data (263K+ samples). Built and optimized an XGBoost model achieving 0.80 ROC-AUC and 0.73 Recall for COVID-19 prediction. DOI: 10.1109/SysCon66367.2026.11503514*
- Bindu Chandra Shekar Reddy.Google Scholar Profile.Full list of research publications available at:
<https://scholar.google.com/citations?user=k67nfawAAAAJ>

Education

California State University, Sacramento

Master of Science, Computer Science

Aug 2024 - May 2026

Sacramento, CA

KNS Institute of Technology (VTU)

Bachelor of Engineering, Computer Science and Engineering

Aug 2018 - Jun 2022

Bengaluru, India